Contents

1 Uniform Asymptotic Expansion for Gegenbauer Polynomials: New Approach 1
Leonid Bakaleinikov and Alexander Silbergleit
1.1 Introduction 1
1.2 Reduction of Differential Equation 2
1.2.1 Transformation of the Equation 2
1.2.2 Chebyshev Polynomials 3
1.3 Asymptotic Anzatz and the Set of Equations for its Coefficients . . . 5
1.3.1 Lowest Order Approximation 5
1.3.2 Asymptotic Anzatz and Equations for Coefficients 5
1.4 Uniform Asymptotic Expansion 7
1.4.1 Finding Coefficients $A_k^{\lambda}(\theta)$ 7
1.4.2 Full Uniformly Valid Expansion 9
1.5 Conclusion 10
1.5.1 Explicit Asymptotic Formula 10
1.5.2 Comparison of Various Uniform Expansions 11
1.5.3 Final Remarks 15
References 16

2 Asymmetric Buckling of Circular and Annular Plates under Normal Pressure 19
Svetlana M. Bauer, Boris N. Semenov, Eva B. Voronkova, and Nikita P. Dorofeev
2.1 Introduction 19
2.2 Problem Formulation 20
2.3 Numerical Results and Discussion 23
2.3.1 Finite Element Modeling 24
2.3.2 Buckling of the Annular Plates 27
2.4 Conclusion 28
References 28

3 Comparison of Hyperelastic Corneal Models for Assessing Eye Deformations After MyoRing Implantation 29
Svetlana M. Bauer, Liudmila A. Venatovskaya, Vladimir V. Kornikov, and Dmitriy V. Matchenko
3.1 Introduction 29
3.2 Problem Statement 30
3.3 Material Models 31
3.4 Results and Discussion 33
3.5 Conclusion 35
References 35

4 The Effect of the Boundary Conditions on Free Vibrations of a String Resting on a Winkler Foundation 37
M. Dykes, Julius Kaplunov, and Danila Prikazchikov
4.1 Introduction 37
4.2 Boundary Value Problem 38
4.3 Properties of the Eigensolutions 40
4.4 Numerical Results 41
4.5 Concluding Remarks 43
References 45

5 Bending Waves in Beams Lying on Generalized Deformable Foundations: Elastic, Viscoelastic and Nonlinear-Elastic Ones 25
Vladimir I. Erofeev, Alexey O. Lenin, Elena E. Lisenkova, and Ivan S. Tsarev

6 Operator Algebra in Linear Viscoelasticity: Experimental Aspects . . 27
Dimitri V. Georgievskii

7 Surface Shear Horizontally Wave Propagation in Half-Space Periodically Stratified with Interfaces of Imperfect Elastic Contact . . 29
K.B. Ghazaryan, R.A. Ghazaryan, and S.A. Terzyan

8 Dynamics of a Vortex in a Regulable Rectangular Channel 31
Vasily Govorukhin and Boris Goncharov

9 On the Plane Contact Problem of a Punch with a Previously Unknown Contact Area in the Presence of Static Friction 33
Vazganush N. Hakobyan, Lilit L. Dashtoyan, Harutyun A. Amirjanyan, and Lusine V. Hakobyan

10 On the Plane Contact Problem of a Punch with a Previously Unknown Contact Area in the Presence of Static Friction 35
Vahram N. Hakobyan, A.V. Gasparyan, and A.V. Sahakyan

11 Magneto-Electric Coefficients of Thermo-Magneto-Electro-Elastic Multilayer Composites: Effective Properties for Nonperfect Connectivity Between Layers 37
J. D. Hasanyan and D. J. Hasanyan

12 Influence of the Moisture Desorption Process on the Modulus of Elasticity and Strength of Lightweight Cement Concrete and Cement Soil Composite 39
K.A. Karapetyan

13 An Asymptotic Model for the Nonstationary Waves in the Shells of Revolution Initiated by the LT type Edge Shock Loading 41
I. V. Kirillova and L. Yu. Kossovich

14 Modeling of Convective Flows in a Porous Circular Enclosure 43
Pavel Kokhanov and Vyacheslav Tsybulin

15 Effect of Nanosized Surface Roughness in Solids with the Account of Steigmann–Ogden Mode 45
Sergey Kostyrko, Mikhail Grekov, and Holm Altenbach

16 Eigenvalues of a Lightly Loaded Lubricated Oscillating Contact 47
Ilya I. Kudish and Sergei S. Volkov

17 Microstructure and Micro- and Nano-Properties of Al-Cr-B-N Coatings 49
Vasilina A. Lapitskaya, Tatyana A. Kuznetsova, Anastasiya V. Khabarava, Sergei A. Chizhik, Bogdan Warcholinski, Adam Gilewicz, Andrey L. Nikolaev, and Sergei M. Aizikovich

18 On Model of Coupled Thermomechanics and Electrodynamics 51
S. Lurie and P. Belov

19 Investigation of the Effect of Magnetostriction in a Rectangular Dielectric Plate Under the Longitudinal Magnetic Field 53
M.A. Mikilyan

20 Exact Solution of a Class of Integral Equations Describing the Problems of Contact Interaction of Stringers with Massive Elastic Bodies 55
S.M. Mkhitaryan

21 Discrimination of Propagating Weak Discontinuities Surfaces in Semi-Isotropic Micropolar Thermoelastic Media 57
Evgenii V. Murashkin

22 Detachment of the elastic half-plane from the punch attached to it . . 59
V. I. Ostryk

23 The Problem of Interaction of Longitudinal Shear Waves with thin Defects in the Shape of a Broken Line 61
Vsevolod Popov

24 Polarization Crosses of Transverse Plane Waves in Micropolar Elastic Media 63
Yuri N. Radayev

25 Fuzzy-Set Methods of Parametric Uncertainty Accounting in the Investigation of Stability Thin-Walled Structures Models 65
Valeriy I. Storozhev and Sergey V. Storozhev

26 Development of the Boundary Equations Method for Wave Problems in 2D Domains with Irregularities of the Boundary Curve . 67
Mezhlum Sumbatyan and Evgeny Titarenko

27 Increasing the Natural Frequency of Oscillations in FGM Rods 69
Alexander O. Vatulyan and V.O. Yurov

28 On the Stress State of a Poroelastic Hollow Finite Cylinder 71
Natalya Vaysfeld and Zinaida Zhuravlova

29 The Boundary Value Problem of the Behavior of the Transverse Electromagnetic Wave 73
Michael Zhukov and Tatyana Dolgikh

30 Mechanics of Transformable Large-Scale Space Structures with Shape Memory Actuator 75
Vladimir N. Zimin, Alexey V. Krylov, and Artur O. Shakhverdov
30.1 Introduction 76
30.2 Experimental Studies of the Active Element Made of a Material with SME 77
30.3 Experimental Study of the Deformation and Force Characteristics of an Active Element made of a Material with SME 77
30.3.1 Tests with Spring Elements 78
30.3.2 Tests with Weights 79
30.4 Experimental Studies of the Functional Characteristics of an Active Element made of a Material with SME 81
30.5 Experimental Studies of the Reduction of Significant Linear Dimensions of an Active Element made of a Material with SME . . 82
30.6 Model of Actuator's Operation 83
30.7 Discussion of the Results 89
30.8 Conclusion 90
References 91

Chapter 2
Asymmetric Buckling of Circular and Annular Plates under Normal Pressure

Svetlana M. Bauer[0000−0003−3732−2110], Boris N. Semenov[0000−0002−8870−6520], Eva B. Voronkova[0000−0002−3065−4473], and Nikita P. Dorofeev[0000−0002−9094−7956]

Abstract Bifurcation of axisymmetric equilibrium states of inhomogeneous annular and circular plates subjected to internal pressure is studied. The effect of material heterogeneity and ratio of inner to outer radii on the buckling load is examined. Finite element and analytical approaches are discussed. It is shown that if the elasticity modulus decreases away from the center of a plate, the critical pressure for unsymmetric buckling is sufficiently lower than for a plate with constant mechanical properties. For solid plates - whether homogeneous or non-uniform and weakened toward the edge - the results of obtained by both methods are closed to each other The calculations of the critical load for annular plates, conducted using the ANSYS finite element package and shell elements, show qualitative agreement with the analytical solutions of the problem.

2.1 Introduction

Circular and annular plates are of considerable significance across a range of applications and they often exhibit large deflections. Symmetrical instability of the thin circular plates subjected to transverse pressure was first analyzed by Panov and Feodos'ev in [1]. They approximated the prebuckling axisymmetric state by a function with only one unknown parameter and examined the unsymmetrical problem by variational method. This approach led to incorrectly determining the critical pressure and buckling mode.

Later, Feodos'ev showed that the elastic surface of plates or shells under large deformations could not be described by one or two unknown parameters in approxi-

Svetlana M. Bauer · Boris N. Semenov · Eva B. Voronkova · Nikita P. Dorofeev
St. Petersburg State University, 7/9 Universitetskaya Emb., St. Petersburg, 199034, Russian Federation,
e-mail: s.bauer@spbu.ru, b.semenov@spbu.ru, e.voronkova@spbu.ru, st067926@student.spbu.ru

mating functions [2]. The existence of unsymmetric equilibrium states for a circular plate under a symmetrical load was shown by Morozov [3, 4].

Cheo and Reiss conducted an analytical study on the axially unsymmetric equilibrium states of circular plates subjected to surface load [5]. They pointed out that Panov and Feodos'ev incorrectly identified an unsymmetrical state and underlined the necessity to accurately calculate the plate's axisymmetrical state before buckling more to obtain correct values for buckling pressure and ~~mode number~~.

Many authors have discussed the possibility of unsymmetrical buckling of internally pressurised circular plates or shallow spherical shells. Manuylov and his coauthors investigated the problem of wrinkling in circular plates using the finite element method [6–8]. They observed that the deformed plate profile identified by Panov in [1] corresponded to a clamped plate, while in the case of moving support, the plate adopts a "bowl" shape under normal pressure. In the experimental study [9] on uniform heating of thin circular plates with fixed edges, the authors also referred to Panov and Feodos'ev results when discussing the formation of waves near the plate edge.

The wrinkling of a uniformly stretched circular plate under transverse pressure was investigated by Coman in [10]. He used numerical and asymptotical approaches to find the buckling pressure and wave number. The plate material was assumed to be isotopic and Föppl–von Kármán (FvK) theory for shallow shells and plates [11, 12] was applied to describe the axisymmetrical equilibrium state.

The unsymmetrical buckling of radially non-uniform orthotropic circular plates was discussed in [13–16]. The deformation before buckling was obtained using the shallow shell theory. The influence of edge fixing conditions on the magnitude of the critical load and the form of stability loss was also discussed. In [16], the instability of radially non-uniform annular plates was also considered. In the papers, analytical solutions were always derived using shallow shell equations.

This paper compares the results obtained in the analytical solution using the shallow shell theory with those obtained by finite element modelling in the problem of stability of radially non-uniform solid and annular plates.

2.2 Problem Formulation

We consider a thin circular plate of radius R and uniform thickness h. The plate is subjected to uniformly distributed pressure P as shown in Fig. 2.1. The plate is made

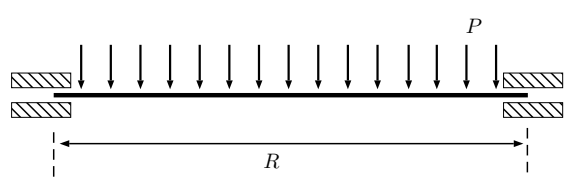


Fig. 2.1 ~~Distribution of displacements in z-direction for the homogeneous plate with 14 waves in the circumferential direction. The load is 6 MPa.~~

of elastic material with Young's modulus $E(r)$ varying in the radial direction and Poisson's ratio ν. The polar coordinate system is adopted here.

The behaviour of the plate with spatial inhomogeneity undergoing large deflections can be formulated in terms of the the transverse displacement $w(r,\theta)$ and the stress function $F(r,\theta)$ by the Donnell-Mushtari-Vlasov equations [17–20] can be written in the form [15]

$$\begin{gathered} D\Delta\Delta w + D'(r)L_1^+(w) + D''(r)L_2^+(w) = P + L(w,F), \\ \frac{\Delta\Delta F}{E} + \left(\frac{1}{E}\right)' L_1^-(F) + \left(\frac{1}{E}\right)'' L_2^-(F) = -\frac{h}{2}L(w,w), \end{gathered} \tag{2.1}$$

where r, θ are the polar coordinates,

$$D(r) = \frac{E(r)h^3}{12(1-\nu^2)}$$

is the bending stiffness and $E(r)$ describes varying Young's modulus of the plate. Δ is the usual Laplacian in polar coordinates and

$$L(x,y) = x''\left(\frac{y'}{r} + \frac{\ddot{y}}{r^2}\right) + y''\left(\frac{x'}{r} + \frac{\ddot{x}}{r^2}\right) - 2\left(\frac{\dot{x}}{r}\right)'\left(\frac{\dot{y}}{r}\right)',$$

$$L_1^{\pm}(y) = 2y''' + (2\pm\nu)\frac{y''}{r} + 2\frac{(\ddot{y})'}{r^2} - \frac{y'}{r^2} - 3\frac{\ddot{y}}{r^3}, \quad L_2^{\pm}(y) = y'' \pm \nu\left(\frac{y'}{r} + \frac{\ddot{y}}{r^2}\right).$$

In writing the above, a prime and dot denote a derivative with respect to the radial r and circumferential θ coordinates, respectively.

The outer edge of the plate is clamped in transverse direction, but the boundary can move freely in the radial direction; therefore, at the edge $r = R$, we have $w = w' = T_{rr} = T_{r\theta} = 0$. The stress-resultants T_{rr}, $T_{\theta\theta}$, $T_{r\theta}$ are expressed through the stress function $F(r,\theta)$ according to

$$T_{rr} = \frac{F'}{r} + \frac{\ddot{F}}{r^2}, \quad T_{r\theta} = F'', \quad T_{r\theta} = -\left(\frac{\dot{F}}{r}\right)'.$$

We introduce the following dimensionless quantities

$$r^* = \frac{r}{R}, \quad w^* = \beta\frac{w}{h}, \quad P^* = \beta^3\frac{PR^4}{E_{\text{av}}h^4}, \quad F^* = \beta^2\frac{F}{E_{\text{av}}h^3}, \quad \beta^2 = 12(1-\nu^2). \tag{2.2}$$

Here E_{av} is an average value of Young's modulus

$$E_{\text{av}} = \frac{1}{S}\iint\limits_S E(r)r\,dr\,d\theta, \quad E(r) = E_0 f(r),$$

where $f(r)$ is a smooth position function and S denotes the area of the plate mid-surface.

The dimensionless form of the system (2.1) is

$$\begin{gathered} g_1(r)\Delta\Delta w + g_1'(r)L_1^+(w) + g_1''(r)L_2^+(w) = P + L(w,F), \\ g_2(r)\Delta\Delta F + g_2'(r)L_1^-(F) + g_2''(r)L_2^-(F) = -L(w,w)/2, \\ g_1(r) = E_0 f(r)/E_{av}, \quad g_2(r) = 1/g_1(r). \end{gathered} \tag{2.3}$$

To find the bifurcation equation, we consider perturbations of the form

$$w(r,\theta) = w_0(r) + \varepsilon w_n \cos(n\theta), \quad F(r,\theta) = F_0(r) + \varepsilon F_n \cos(n\theta), \tag{2.4}$$

where $w_0(r)$, $F_0(r)$ describe prebuckling axisymmetric state, ε is an infinitesimal parameter, n is the ~~mode number,~~ and $w_n(r)$, $F_n(r)$ are non-symmetrical components.

The functions $w_0(r)$, $F_0(r)$ can be found from the equations governing the axisymmetic behavior of the plate are

$$\begin{gathered} g_1\left(\Theta_0'' + \frac{\Theta_0'}{r} - \frac{\Theta_0}{r^2}\right) + g_1'\left(\Theta_0' + \nu\frac{\Theta_0}{r}\right) = \frac{Pr}{2} + \frac{\Theta_0\Phi_0}{r}, \\ g_2\left(\Phi_0'' + \frac{\Phi_0'}{r} - \frac{\Phi_0}{r^2}\right) + g_2'\left(\Phi_0' - \nu\frac{\Phi_0}{r}\right) = -\frac{\Theta_0^2}{2r}, \end{gathered} \tag{2.5}$$

where $\Theta_0 = \Theta_0(r) = w'(r)$ and $\Phi_0 = \Phi_0(r) = F'(r)$.

In axisymmetric case the boundary conditions at the edge $r = 1$ of the plate are

$$\Theta_0(1) = \Phi_0(1) = 0. \tag{2.6}$$

A the plate center, we have the regularity conditions

$$\Theta_0(0) = \Phi_0(0) = 0. \tag{2.7}$$

The bifurcation equations are obtained after substitution of (2.4) in (2.3), followed by the linearization procedure with respect to ε. The equations governing the asymmetric state of the plate are

$$\begin{aligned} g_1\Delta_n\Delta_n w_n + \mathcal{L}_1(g_1, w_n) &= \frac{w_n''}{r}\Phi_0 + \frac{F_n''}{r}\Theta_0 \\ &\quad + \Theta_0'\left(\frac{F_n'}{r} - \frac{n^2}{r^2}F_n\right) - \Phi_0'\left(\frac{w_n'}{r} - \frac{n^2}{r^2}w_n\right), \\ g_2\Delta_n\Delta_n F_n + \mathcal{L}_2(g_2, F_n) &= -\frac{w_n''}{r}\Theta_0 - \Theta_0'\left(\frac{w_n'}{r} - \frac{n^2}{r^2}w_n\right), \end{aligned} \tag{2.8}$$

where $\Delta_n = d^2/dr^2 + r^{-1}d/dr - n^2/r^2$ and the operators $\mathcal{L}_1$, $\mathcal{L}_2$ are defined as

$$\mathcal{L}_1(g_1, w_n) = g_1' L_{1n}^{+}(w_n) + g_1'' L_{2n}^{+}(w_n),$$
$$\mathcal{L}_2(g_2, F_n) = g_2' L_{1n}^{-}(F_n) + g_2'' L_{2n}^{-}(F_n)$$

with

$$L_{1n}^{\pm}(y) = 2y''' + \frac{2 \pm \nu}{r} y'' - \frac{2n^2 + 1}{r^2} \ddot{y} + \frac{3n^2}{r^3} \ddot{y},$$
$$L_{2n}^{\pm}(y) = y'' \pm \nu \left(\frac{y'}{r} - \frac{n^2}{r^2} \ddot{y} \right).$$

The boundary conditions are reduced to

$$\begin{aligned} &F_n'(1) - n^2 F_n(1) = F_n'(1) - F_n(1) = w_n(1) = w_n'(1) = 0, \\ &w_n'(0) = F_n'(0) = 0. \end{aligned} \tag{2.9}$$

2.3 Numerical Results and Discussion

To solve nonlinear axisymmetric problem (2.5) together with the boundary conditions (2.6)-(2.7) we use programming language Python. The critical bifurcation load is the smallest value of P for which Eqs. (2.8) with boundary conditions (2.9) have non-trivial solutions for any integer n. We calculate critical values of P numerically by finite difference method [5, 14].

For a homogeneous circular we obtain the value of a buckling load of axisymmetric equilibrium state $P^{\mathrm{cr}} = P_0^{\mathrm{cr}} = 64956$. We consider two laws for material inhomogeneity, linear

$$E = E_0(1 + qr)$$

and exponential

$$\cancel{E = E_0 e^{-qr}},$$

to study the effect of the varying rate of inhomogeneity on the critical load and buckling mode. The buckling load for unsymmetrical buckling was calculated numerically over a large range of parameters E_0, q, but for constant average value of the elastic modulus. The results are summarized in Tables 2.1 and 2.2. The parameter value $q = 0$ corresponds to uniform plate with constant Young's modulus.

Table 2.1: Normalized buckling load ($P^{\mathrm{cr}}/P_0^{\mathrm{cr}}$) for the heterogenous plate with linear law for material inhomogeneity, $E = E_0(1 + qr)$.

	$q = 0$	$q = -0.3$	$q = -0.5$	$q = -0.7$	$q = -0.9$
$P^{\mathrm{cr}}/P_0^{\mathrm{cr}}$	1	0.9	0.8	0.68	0.48
~~Mode number~~ m	14	14	14	14	15

Table 2.2: Normalized buckling load (P^{cr}/P_0^{cr}) for the heterogenous plate with exponential law for material inhomogeneity, $\cancel{E = E_0 e^{-qr}}$.

	$q = -1$	$q = -2$	$q = -2.5$	$q = -3$	$q = -4$	$q = -5$
P^{cr}/P_0^{cr}	0.75	0.55	0.48	0.4	0.28	0.18
~~Mode number~~, m	14	14	14	15	16	17

For the heterogeneous plate with a weakened edge, the normalized buckling pressure P^{cr}/P_0^{cr} decreases as the rate of inhomogeneity $|q|$ increases, whereas the buckling mode m increases with $|q|$. If the stiffness of the plate increases from the center to the edge ($q > 0$), the critical load also rises, while the ~~mode number~~ remains equal to 14. Similar results we reported in [13, 15].

It is important to note that the results for the linear law with ($q = -0.9$) and the exponential law with ($q = -2.5$) are strikingly similar. In both cases, Young's modulus near the center of the plate is approximately ten times greater than that at the edge. This finding suggests that the critical load is influenced not only by the mechanical properties at the edge but also by the average value of the Young modulus across the entire plate.

2.3.1 Finite Element Modeling

We used the finite element method to study the stability of the symmetrical equilibrium state of a circular plate in the applied software package ANSYS. We consider a circular plate with of radius $R = 0.5$ m, thickness $\cancel{h = 5 \cdot 10^{-3}\ \text{m}}$. The plate lies in the plane $z = 0$ and fills a region

$$0 \leqslant \sqrt{x^2 + y^2} \leqslant 0.5.$$

The plate is divided into 8267 elements of SHELL181 type with radial condensation of a mesh to $R_{in} = 0.1$ m. The mesh is regular in the region $R_{in} \leqslant r \leqslant R$ and irregular near the center of the plate when $0 \leqslant r \leqslant R_{in}$. At the boundary of the plate $r = R$ we assume a sliding support conditions with displacement $u_z = 0$ and $\partial u_z/\partial r = 0$.

To eliminate movements of the plate as a rigid body in the plane $z = 0$ we restrained the movements of the plate center and the point on the boundary. The point with coordinates $x = 0, y = 0$ (the plate center) can only move along z-axis during deformation, and the point on the plate boundary ($x = -0.5, y = 0$) — only along the x-axis. The plate is subjected to uniformly distributed pressure over the entire surface.

We use the standard procedure to perform nonlinear stability analysis. The geometric imperfections is introduced in the initial state of the plate, allowing the nodes to deflect from the plane $z = 0$. These deflections is taken as less than 0.2% of the

plate thickness. We solve the nonlinear problem that allows for large deformations of the plate and find the values of the pressure, for which the rapid localized growth of deflections is observed for an insignificant increase in pressure, i.e., it is deformation buckling.

First we study a plate with uniform material properties with Young's modulus $E_0 = 205$ GPa and Poisson's ratio $\nu = 0.3$. For homogeneous plate unsymmetrical buckling occurs at $P_{\mathrm{cr}} = 3.6$ MPa with 14 waves in a buckling mode. This value corresponds to a dimensionless quantity of 63370, only 3% smaller than the results obtained using the finite difference method described in the previous section (64956). We show the distribution of normal displacements at the higher load of 6 MPa in Fig. 2.2 for the sake of illustration of waves formation.

For the plate with varying mechanical properties, we consider two cases: the Young modulus changes linearly

$$E = E_0 \left(1 - 0.9\frac{r}{R}\right)$$

or exponentially with

$$E = E_0 \exp\left(-2.5\frac{r}{R}\right)$$

in the radial direction. Here R is the radius of the plate. When the Young modulus changes linearly the plate buckles at pressure $P^{\mathrm{cr}} = 0.5$ MPa with $n = 14$ waves is the circumferential directions (Fig. 2.3). If Young's modulus varies exponentially, unsymmetrical wrinkling starts to appear at a lower value of $P^{\mathrm{cr}} = 0.365$ MPa with formation of $n = 14$ waves in the circumferential directions (Fig. 2.4). The buckling pressure in the two analyzed cases varies differs due to variations in the average elastic modulus. For both linear and exponential law, Young's modulus at the edge of the plate is approximately ten times lower than that near the center. However, in the linear case, the average elastic modulus is 82 GPa, which is nearly twice as large as the average elastic modulus for the exponential case, calculated at 47 GPa.

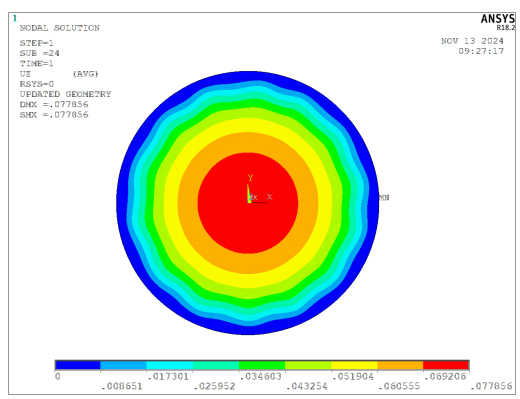
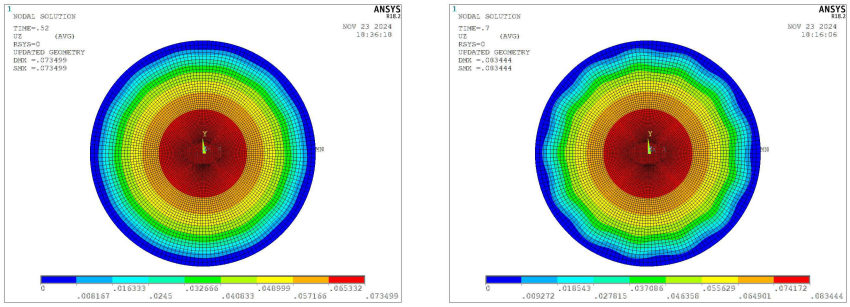
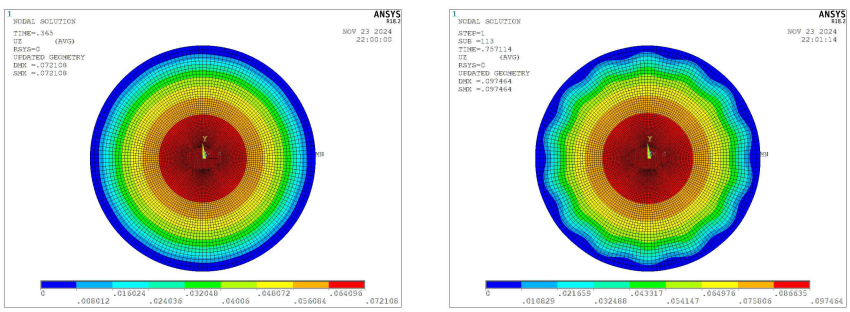


Fig. 2.2 Distribution of displacements in z-direction for the homogeneous plate with 14 waves in the circumferential direction. The load is 6 MPa.



(a) **(b)**

Fig. 2.3: Distribution of displacements in z-direction for the heterogenous plate with linear law of material inhomogeneity ($q = -0.9$): **(a)** Deformation right after the buckling into unsymmetrical state. The pressure is $p = 0.52$ MPa. The buckling mode $n = 14$. **(a)** Deformation right under the pressure of $p = 0.7$ MPa with 14 waves in the circumferential direction.



(a) **(b)**

Fig. 2.4: Distribution of displacements in z-direction for the heterogenous plate with exponential law of material inhomogeneity ($q = -2.5$): **(a)** Deformation right after the buckling into unsymmetrical state. The pressure is $p = 0.365$ MPa. The buckling mode $n = 14$. **(b)** Deformation right under the pressure of $p = 0.76$ MPa with 14 waves in the circumferention direction.

To compare the outcomes of the FEM and analytical approaches, we conducted calculations based on two different models of variation in the elastic modulus, intentionally not maintaining a constant average value for the elastic modulus. he dimensional critical loads determined through the analytical method are 0.7 MPa for the linear law and 0.4 MPa for the exponential law. These results are in good agreement with the FEM calculations.

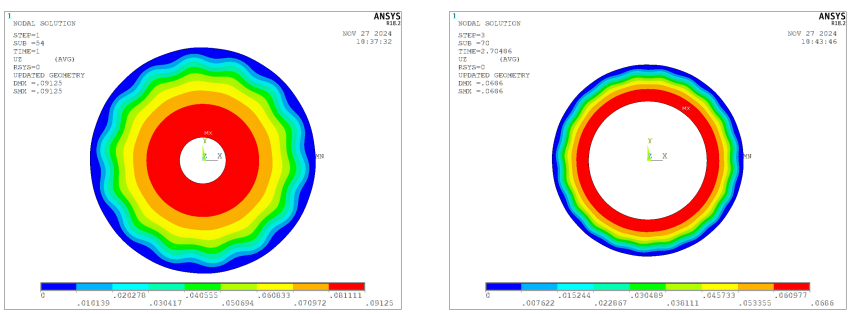
Manuylov and coauthors considered a circular plate with the same radius and thickness as in the present paper [7]. They reported that the plate made of structural steel buckles in the unsymmetrical state at load $q_{\text{cr}} = 3.57$ MPa with the formation

of 13 waves in the circumferential direction. ~~The displacement at the plate centre equals 13.8h or 6.9 cm.~~

2.3.2 Buckling of the Annular Plates

Axisymmetrical stability of an annular plates with uniform mechanical properties was also studied with finite elements methods. We assume that the inner edge of the plate $r = R_{\text{in}}$ is supported by roller that can slide along a vertical wall: $u_r = u_\theta = \partial u_z/\partial r = 0$. The buckling pressure increases as the radius of the opening R_{in}/R increases together with the buckling mode n. The plate with $R_{\text{in}} = 0.1$ m ($R_{\text{in}}/R = 0.2$) loose axisymmetrical stability at the critical pressure $P^{\text{cr}} = 4.25$ MPa with 14 waves in the circumferential direction. When the radius of the opening increases to $R_{\text{in}} = 0.3$ m ($R_{\text{in}}/R = 0.6$) the annular plate exhibits unsymmetrical behaviour, when the load reaches the value of 20 MPa, and the ~~mode number~~ is 20. For clarity, Fig. 2.5a shows the distribution of normal displacements when the load reaches the value of 10 MPa, and the ~~mode number~~ is 14 for the plate with relatively small opening in the center. For the plate with $R_{\text{in}}/R = 0.6$, the unsymmetrical state with waves formation near the outer edge is depicted at the load of 30 MPa with the ~~mode number~~ 20 (Fig. 2.5b).

In our previous results on the stability of annular plates we report that the buckling mode decreases as the relative radius of R_{in}/R increases. One possible reason for the discrepancy between the current and previous results may be the use of shallow shell theories to describe the symmetrical state before the buckling (Fig. 2.6).



(a) **(b)**

Fig. 2.5: Distribution of displacements in z-direction for the annular plate with the constant elastic nodulus: **(a)** The inner radius is $R_{\text{in}} = 0.1$ m. The pressure is $p = 10$ MPa. The buckling mode $n = 14$. **(b)** The inner radius is $R_{\text{in}} = 0.3$ m. The pressure is $p = 30$ MPa. The buckling mode $n = 20$.

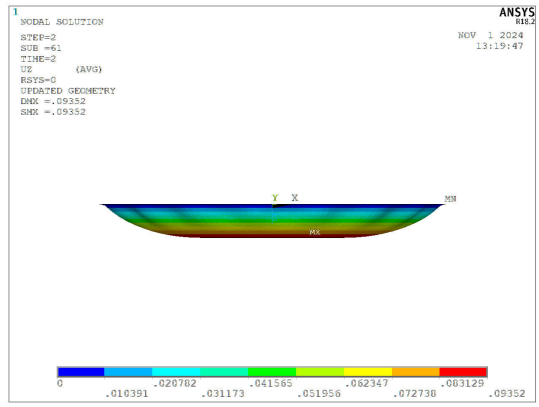


Fig. 2.6 "Bowl"-shaped deformation of an annular plate $R_{\text{in}}/R = 2/5$ after the loss of symmetrical stability

2.4 Conclusion

The calculations of the critical load for annular plates, conducted using the ANSYS finite element package and shell elements, show qualitative agreement with the analytical solutions of the problem.

For solid plates — whether homogeneous or non-uniform and weakened toward the edge — the results are nearly identical. As the inner radius of the annular plate increases, the value of the critical load also rises, and the deflection shape near the outer edge changes.

Furthermore, the discrepancy between the results obtained through the finite element method and those derived from the equations of shallow shells increases with higher loads. This variation occurs because the equations of shallow shells do not accurately represent the subcritical state under these conditions. To address this issue, especially for plates with hinged support at the outer edge, it is essential to employ the equations of the general theory of shells.

Acknowledgements This work was supported by the Russian Science Foundation (no. grant 20-11-20083)

References

[1] Panov D, Feodosiev V (1948) On the equilibrium and loss of stability of shallow shells in the case of large displacement in Russ. Prikladnaya Matematika i Mekhanika 12:389–406

[2] Feodos'ev V (1963) On a method of solution of the nonlinear problems of stability of deformable systems. Journal of Applied Mathematics and Mechanics 27(2):392–404, DOI 10.1016/0021-8928(63)90008-X

[3] Morozov N (1961) On the existence of a non-symmetric solution in the problem of large deflections of a circular plate with a symmetric load (in Rus.). Izvstiya Vysshikh Uchebnykh Zavedenii Matematika (2):126–129
[4] Morozov N (1962) A qualitative investigation of a circular, symmetrically compressed plate under a large edge load (proof of the appearance of crimping) (in Rus.). Doklady Akademii Nauk SSSR 147(6):1318–1319
[5] Cheo L, Reiss E (1973) Unsemmetric wrinkling of circular plates. Quarterly of Applied Mathematics 31(1):75–91
[6] Manuylov GA, Kositsyn SB, Begichev MM (2011) The investigation of stability of elastic plates and shells with the help of fe modeling. Structural Mechanics of Engineering Constructions and Buildings (1):58–65
[7] Manuylov GA, Kosytsyn SB, Begichev MM (2014) About initial imperfection sensitivity of some thin-walled strustures (in Russ.). International Journal for Computational Civil and Structural Engineering 10(1):111–117
[8] Manuylov GA, Kosytsyn SB, Begichev MM (2017) On computational differences of critical points on equilibrium curve (in Russ.). International Journal for Computational Civil and Structural Engineering 13(2):125–135, DOI 10.22337/2587-9618-2017-13-2-125-135
[9] Goldstein RV, Popov AL, Kozintsev VM, Chelyubeev DA (2016) Non-axisymmetric edge buckling of circular plates when heated. PNRPU Mechanics Bulletin (2):45–53, DOI 10.15593/perm.mech/2016.2.04
[10] Coman C (2018) On the asymptotic reduction of a bifurcation equation for edge-buckling instabilities. Acta Mechanica 229(3):1099–1109, DOI 10.1007/s00707-017-2036-8
[11] Föppl A (1907) Vorlesungen über technische Mechanik, vol 5. B.G. Teubner, Leipzig
[12] von Kármán T (1910) Festigkeitsprobleme im maschinenbau. In: Klein F, Müller C (eds) Encyklopdie der Mathematischen Wissenschaften, vol IV/4, B.G. Teubner, Leipzig, pp 311–385, DOI 10.1007/978-3-663-16028-1_5
[13] Bauer S, Voronkova E, Morozov N (2012) On unsymmetrical equillibrium states of circular plates under normal pressure. In: Chigarev A (ed) Theoretical and Applied Mechanics: International Scientific and Technical Collection, Minsk, pp 31–35
[14] Bauer SM, Voronkova EB (2019) Unsymmetrical wrinkling of nonuniform annular plates and spherical caps under internal pressure. In: Altenbach H, Chróścielewski J, Eremeyev VA, Wiśniewski K (eds) Recent Developments in the Theory of Shells, Springer International Publishing, Cham, pp 79–89, DOI 10.1007/978-3-030-17747-8_6
[15] M Bauer SM, Voronkova EB (2020) On buckling behavior of inhomogeneous shallow spherical caps with elastically restrained edge. In: Altenbach H, Chinchaladze N, Kienzler R, Müller WH (eds) Analysis of Shells, Plates, and Beams: A State of the Art Report, Springer International Publishing, Cham, pp 65–74, DOI 10.1007/978-3-030-47491-1_4

[16] Bauer S, Voronkova EB (2024) On asymmetrical equilibrium states of annular plates under normal pressure. Izvestiya of Saratov University Mathematics Mechanics Informatics 24(1):28–34, DOI 10.18500/1816-9791-2024-24-1-28-34
[17] Donnel LH (1976) Nonlinear Theory of Thin Elastic Shells. Engineering Societies Monographs Series New, McGraw-Hill, New York
[18] Mushtari KM, Galimov KZ (1961) Beams, Plates, and Shells. NSF-NASA, Washington
[19] Vlasov VZ (1949) General Theory for Shells and its Application in Engineering (in Russ.). Gostekhizdat, Moscow
[20] Wlassov WS (1958) Allgemeine Schalentheorie und ihre Anwendung in der Technik. Akademie-Verlag, Berlin